



UNSW
SYDNEY

MATH1231 Mathematics 1B – Algebra

Section 11: Set Theory and Probability

Conditional Probability, Independence

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Slides: David Angell, Laure Helme-Guizon and many others

Introduction to probability

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“if I toss ten coins, the probability that seven or more turn up heads is 0.17”.

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If we want to think seriously about probability, these questions need to be addressed – try looking up “measure theory and probability”. For MATH1231, however, we shall be satisfied with a “gut feeling” for probability and will not look very closely into its philosophical basis.

Introduction to statistics

The word **statistics** in common usage means a collection of figures, usually relating to some particular topic: for example, “world population statistics”, or “UNSW student enrolment statistics”.



In mathematical terminology, however, **Statistics** does not mean the numbers themselves but how we get them and what we do with them, “the science of making rational decisions in an uncertain situation”.



Example 2 (food for thought). A new treatment has been developed for cancer. Is it superior to the standard treatment?

The basic idea for answering this question is fairly obvious: treat some patients the old way, some the new way and then compare the results. But there are difficulties. First note that whatever results we obtain, we can never be absolutely certain which treatment is better.

Suppose, for example, that most people undertaking the old treatment are cured, and most taking the new treatment are not. This might mean that the old treatment is better; it might be that patients taking the old treatment were not, before treatment, as badly sick as the others; it might just be a matter of chance – after all, it's not likely that either treatment will cure *everyone*.

....continued

Example 2 (Reflection on Statistics), continued.

- To obtain a sufficiently confident answer, how many patients should we look at? How many should take each treatment? Statistics can advise on **data production**. Note that we use the *science of statistics* before we have any “statistics” (that is, data)!
- We shall generally need to summarise our data, for example, to find the proportion of cures under each treatment. It might be necessary to further subdivide the data according to the patient's age, sex, previous medical history and so on. This is known as **data analysis**. We may need to calculate averages, standard deviations and other statistical measures. If there is a lot of data, tables and graphs will be very useful.
- We shall need to draw conclusions by means of **statistical inference** to answer our questions. This is rarely straightforward! For example, suppose that the old treatment gives 62% cures and the new 63%. Can we definitely say that the new treatment is better? What if the proportions were 62% and 64%? Or 62% and 70%?

Introduction to statistics, continued

Example 2 (Reflection on Statistics), continued.

To make a decision from our data, we proceed as follows.

Suppose that we already have lots of data on the old treatment and we know that it cures 62% of patients.

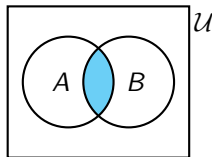
We pick a *significance level*, and by using the techniques of probability derive a statement such as “we will be 95% sure that the new treatment is better if it cures 74% or more of the patients in our sample”.

We then look at the data, see whether or not 74% were cured by the new treatment, and make our decision accordingly.

What's the plan and why?

Theorems and questions of probability are expressed in the language of **sets**.

Therefore, we will start with some
set theory
(section 9.1 in the course notes)



Concept. “A **set** is a collection into a whole, A , of definite and separate objects of our intuition or our thought. These objects are called the **elements** of A .” (Georg Cantor, translated by P.E.B. Jourdain.)

Note. A set must be **well-defined**: it must be possible in principle (though maybe very difficult in practice) to decide whether any particular “object” is an element of the set or not.

Notation. We write $x \in A$ to indicate that an “object” x is an element of a set A , and $x \notin A$ to indicate that it is not.

Definitions

- If every element of the set A is also an element of the set B , then we say that A is a **subset** of B and write $A \subseteq B$.
- The **power set** of a set A , denoted $\mathcal{P}(A)$, is the set of all subsets of A .
- A **universal set** is a set containing all elements of all sets that we will need to talk about in some particular topic.
- The **empty set**, denoted $\emptyset$ or $\{ \}$, is the set with no elements.

Example 3. Consider the set $A = \{ \triangle, \circ, \square \}$



a) The set A has ... elements and ... subsets, with $\mathcal{P}(A) = \dots$

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a) The set A has ... elements and ... subsets, with $\mathcal{P}(A) = \dots$

Answer: The set $A = \{ \triangle, \circ, \square \}$ has three elements and eight subsets, with $\mathcal{P}(A) = \{ \emptyset, \{ \triangle \}, \{ \circ \}, \{ \square \}, \{ \triangle, \circ \}, \{ \triangle, \square \}, \{ \circ, \square \}, \{ \triangle, \circ, \square \} \}$.

b) TRUE or FALSE?

Statement 1 : $\triangle \in A$.

Statement 2 : $\{ \triangle \} \in A$.

Statement 3 : $\{ \triangle \} \subseteq A$.

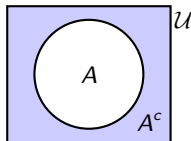
Statement 4 : $\{ \triangle \} \in \mathcal{P}(A)$.

Statement 5 : $\{ \triangle \} \subseteq \mathcal{P}(A)$.

Set theory – Set operations

For these next examples, let $\mathcal{U} = \{1, 2, 3, 4, 5\}$, $S = \{1, 3, 5\}$, $T = \{2, 3, 5\}$.

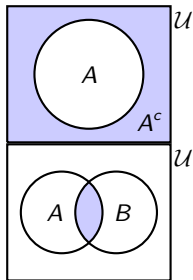
- The **complement** of a set A , written A^c , is the set of all elements in the universal set that are not in A . That is, $A^c = \{x \in \mathcal{U} : x \notin A\}$.
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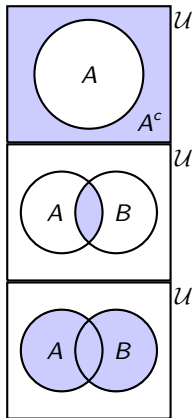
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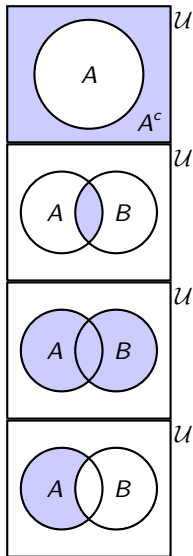
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- The **difference** of two sets A and B , written $A - B$, is the set of all elements in A which are not in B . That is, $A - B = \{x \in \mathcal{U} : x \in A \text{ and } x \notin B\}$.
e.g. $S - T = \{1\}$ and $T - S = \{2\}$.

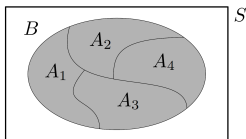


Definitions (disjoint, pairwise disjoint, partition)

- Sets $A_1, A_2, \dots, A_k$ are said to be **disjoint** or **mutually exclusive** if there are no elements common to all of the sets, that is, if $A_1 \cap A_2 \cap \dots \cap A_k = \emptyset$.
- Sets $A_1, A_2, \dots, A_k$ are **pairwise disjoint** if there are no elements common to any two of the sets, that is, if any two of them are disjoint.
- Sets $A_1, A_2, \dots, A_k$ are said to **partition** a set B if $A_1, A_2, \dots, A_k$ are pairwise disjoint and $A_1 \cup A_2 \cup \dots \cup A_k = B$.

Example 5.

- a) The sets A_1, A_2, A_3, A_4 partition B . b) Find a partition of $S = \{2, 3, 5, 7\}$



- c) Consider the sets $A_1 = \{1, 4, 6, 9\}$, $A_2 = \{1, 6, 7\}$, $A_3 = \{2, 5, 7\}$.



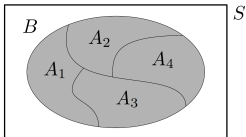
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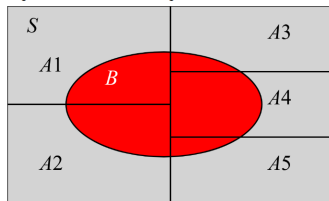
Answer: They are disjoint, but they are not *pairwise* disjoint
(Note that 'pairwise disjoint' $\implies$ 'disjoint').

How to get a partition of a subset if you know a partition of the whole set?

Lemma (Induced partitions)

If $A_1, A_2, \dots, A_k$ partition the set S and if B is a subset of S , then $A_1 \cap B, A_2 \cap B, \dots, A_k \cap B$ partition B .

A partition of S induces a partition of B



source of diagram: <https://www.randomservices.org/random/foundations/Sets.html>

Proof. We have to show that under the given conditions, the sets $A_1 \cap B, A_2 \cap B, \dots, A_k \cap B$ are pairwise disjoint and their union is B .

Firstly, suppose that an element x is in two of these sets, say $A_i \cap B$ and $A_j \cap B$, where $i \neq j$. Then x is in both A_i and A_j ; and this is impossible since the sets $A_1, A_2, \dots, A_k$ are pairwise disjoint.

Secondly, it is clear that each set $A_i \cap B$ is a subset of B , so the union of all these sets is a subset of B . Conversely, any $x \in B$ is also in S ; so $x \in A_i$ for some i ; and hence $x \in A_i \cap B$. Therefore the union of all $A_i \cap B$ is equal to B , and the proof is complete.

Set identities

The set operations, particularly union, intersection and complement, satisfy many identities, including:

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Exercise 6. (in your own time) Check these identities, perhaps with the aid of a Venn diagram.

Sets and counting

Definition (cardinality)

If A is a finite set then $|A|$ denotes the **cardinality** of A , that is, the number of elements in A .

Lemma (The Inclusion–Exclusion Principle)

For any finite sets A, B, C , we have

- $|A \cup B| = |A| + |B| - |A \cap B|$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Comment. There are similar formulae involving any finite number of finite sets.



Illustration for 2 sets

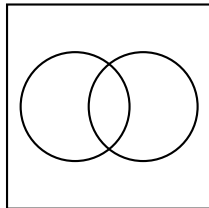
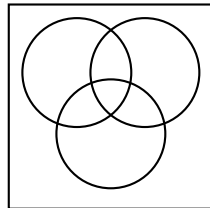


Illustration for 3 sets



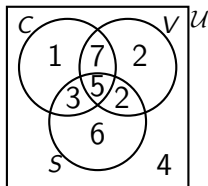
Inclusion-exclusion principle, an example

Example 7. 30 people were asked to select their preferred ice-cream flavours from a list, where more than one selection was allowed. 16 people selected chocolate, 16 selected vanilla, and 16 selected strawberry. 12 people selected both chocolate and vanilla, 8 people selected both chocolate and strawberry, and 7 people selected both vanilla and strawberry. 5 people selected all three flavours. How many surveyed people selected none of the three flavours?

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Solution. Let C , V , and S be the set of people who selected chocolate, vanilla, and strawberry respectively. We can solve this by filling out a Venn diagram:

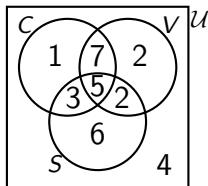


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Alternative solution. Using the inclusion-exclusion principle, we have

$$\begin{aligned}|C \cup V \cup S| &= |C| + |V| + |S| - |C \cap V| - |C \cap S| - |V \cap S| + |C \cap V \cap S| \\ &= 16 + 16 + 16 - 12 - 8 - 7 + 5 = 26,\end{aligned}$$

so the number of people who chose none of the three flavours is $30 - 26 = 4$.

Definition (countable)

A set A is said to be **countable** if its elements can be listed in a (finite or infinite) sequence.

Example 8.

- Any finite set is countable.
- The set of all integers is countable.

$\dots \quad -4 \quad -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad \dots$

- Amazingly, the set of all positive rational numbers is countable!

$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\dots$
$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$	$\dots$
$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$	$\dots$
$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$	$\dots$
$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

- Amazingly, the set of real numbers are *not* countable! This is shown by an ingenious proof known as **Cantor's diagonal argument** (look it up).

Now that we have set theory in place, we can move on to

Probability

Probability. There are many situations in real life where the outcome of our actions, or other people's actions, or natural events, cannot be predicted with certainty. For example,

- tossing a coin and observing which face turns up;
- tossing a coin three times and observing the sequence of results;
- tossing a coin three times and observing how many heads turn up;
- investing \$1000 on the stock market and seeing how much it is worth after a year;
- choosing a person at random from the Australian population and noting their age at their most recent birthday.

Any situation like this is referred to as an **experiment**; this does not necessarily mean the sort of “controlled experiment” that we see in many sciences.

Definition (sample space)

A **sample space** of an experiment is a set of all possible outcomes.

Example 9. Consider the experiment of tossing a coin three times.

- If we are interested in the sequence of results, then the sample space would be

$$\{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \} .$$

- If we are interested in the total number of heads, the sample space would be $\{ 0, 1, 2, 3 \}$.
- If we are only interested the result of the last throw, the sample space would be $\{ H, T \}$.

Take-home message. *There are many possible sample spaces. The choice of a sample space depends on the question you want to answer.*

Events

Often, especially when we come to probability, we are interested in results that do not consist of individual outcomes.

Definition (**event**)

An **event** is a subset of a sample space.

Example 10.

1. In the experiment of tossing a coin three times, and observing the sequence of results, here are some events that might be of interest:
 - (a) the result is two heads followed by a tail: $\{HHT\}$;
 - (b) there are more tails than heads: $\{TTT, HTT, THT, TTH\}$;
 - (c) there is at least one head in the first two tosses
 $\{HHH, HTH, THH, HHT, HTT, THT\}$;
 - (d) there are four or more heads: $\emptyset$.

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Often, especially when we come to probability, we are interested in results that do not consist of individual outcomes.

Definition (event)

An **event** is a subset of a sample space.

Example 10.

1. In the experiment of tossing a coin three times, and observing the sequence of results, here are some events that might be of interest:
 - (a) the result is two heads followed by a tail: $\{HHT\}$;
 - (b) there are more tails than heads: $\{TTT, HTT, THT, TTH\}$;
 - (c) there is at least one head in the first two tosses
 $\{HHH, HTH, THH, HHT, HTT, THT\}$;
 - (d) there are four or more heads: $\emptyset$.
2. In the experiment of observing a person's age last birthday, possible events of interest include
 - (a) the person is over 100 years old: $\{100, 101, 102, \dots\}$;
 - (b) the person is a child under a year old: $\{0\}$;
 - (c) the person is not yet 18 years old: $\{0, 1, 2, \dots, 17\}$.

Probability

We have seen examples where individual outcomes have equal probability, and examples where they do not. Rather than try to analyse in advance what sorts of probabilities may be encountered in practice, it is more convenient to frame a definition which encapsulates the essential features of probability.

Definition (probability)

A **probability** on a sample space S is a function $\mathbb{P}$ from the power set $\mathcal{P}(S)$ to $\mathbb{R}$ which has the following properties.

1. For all $A \subseteq S$ we have $0 \leq \mathbb{P}(A) \leq 1$.
2. $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(S) = 1$.
3. **Additivity property for disjoint events.** If A and B are disjoint subsets of S then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Basic but useful properties of probability

Many other important properties about probability can be deduced from the definition. Here are some of the most fundamental.

Fact (Basic properties of probability)

Let A and B and $A_1, A_2, \dots, A_k$ be events in a sample space S .

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. If $A_1, A_2, \dots, A_k$ are pairwise disjoint, then
$$\mathbb{P}(A_1 \cup A_2 \cup \dots \cup A_k) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots + \mathbb{P}(A_k) .$$
3. **Additivity property.** $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
4. If $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$.



*A bit of space to discuss the ideas behind the **proof**.*

Basic properties of probability, proof

Now that we have the ideas, we can organise them and connect them by sentences to get:

Proof.

[1] The events A and A^c are disjoint, so the additivity property for disjoint events (part 3 of the definition), followed by the second equality in part 2, shows that

$$\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(A \cup A^c) = \mathbb{P}(S) = 1.$$

Our first claim follows easily.

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[2] The second is left as an exercise (use the additivity property for disjoint events and induction).

[3] To prove the third result, note that $A - B$ and $A \cap B$ and $B - A$ are pairwise disjoint; applying this in various ways shows that

$$\mathbb{P}(A \cup B) = \mathbb{P}(A - B) + \mathbb{P}(A \cap B) + \mathbb{P}(B - A)$$

and

$$\mathbb{P}(A) = \mathbb{P}(A - B) + \mathbb{P}(A \cap B) \quad \text{and} \quad \mathbb{P}(B) = \mathbb{P}(B - A) + \mathbb{P}(A \cap B) .$$

Eliminating $\mathbb{P}(A - B)$ and $\mathbb{P}(B - A)$ from these three equations gives the desired result.

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Eliminating $\mathbb{P}(A - B)$ and $\mathbb{P}(B - A)$ from these three equations gives the desired result.

[4] To prove the fourth property, note that if $A \subseteq B$ then $A \cap B = A$; so the third of the above equations can be written

$$\mathbb{P}(B) = \mathbb{P}(B - A) + \mathbb{P}(A) ,$$

and we have

$$\mathbb{P}(A) = \mathbb{P}(B) - \mathbb{P}(B - A) \leq \mathbb{P}(B) .$$

□



Example 11. The probability that a randomly chosen MATH1231 student uses the UNSW help resources (staff consultations, drop-in centre, forums, PASS, ...etc) is 0.4, and the probability that a randomly chosen MATH1231 student gets help from peers (Discord groups, study groups...etc) is 0.5. The probability that they do not use any help, neither from UNSW nor from peers is 0.3. What is the probability that a randomly chosen MATH1231 student uses both the UNSW help resources and the help from peers?

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What is the probability that a randomly chosen MATH1231 student uses both the UNSW help resources and the help from peers?

Solution. Let A be the event that a randomly chosen MATH1231 student uses the UNSW help resources and B be the event that that a randomly chosen MATH1231 student gets help from peers. Then the event that they get help from neither is $(A \cup B)^c$, so by the complement rule we know that

$$P(A \cup B) = 1 - P((A \cup B)^c) = 1 - 0.3 = 0.7.$$

The event that a student uses help both from the UNSW resources and from peers is $A \cap B$. By the addition rule, we know $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, so rearranging gives

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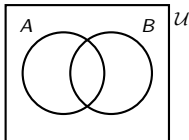
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TIP!

Notice that we can also solve this sort of problem with a Venn diagram representing probabilities instead of cardinalities:



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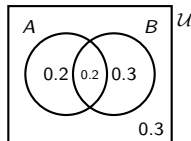
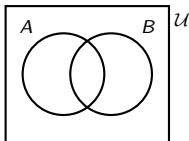
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Further results

We can use the preceding results to prove the following properties (exercise!).

Fact (Further properties of probability)

Let A be an event in a sample space S .

- If S is finite and $A \subseteq S$, then $\mathbb{P}(A) = \sum_{a \in A} \mathbb{P}(\{a\})$.
- If S is finite, then $\sum_{s \in S} \mathbb{P}(\{s\}) = 1$.

Comments. The above results are also true if S is countable. This raises some new issues (the sums involved may be infinite series) and we shall not address this case in the present course.

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Fact (Equally likely outcomes)

If S is finite and $\mathbb{P}(\{s\})$ has the same value for each $s \in S$, we say that **all outcomes are equally likely**. In that situation,

the probability of any individual outcome $s \in S$ is $\mathbb{P}(\{s\}) = \frac{1}{|S|}$

and the probability of any event $A \subseteq S$ is $\mathbb{P}(A) = \frac{|A|}{|S|}$.

Examples of calculation of probabilities

We return to the examples on page 17.

Example 12.

- 1) In tossing three coins, it seems reasonable to assume that heads and tails are equally likely on each toss (if not, say we are tossing three *fair* coins where this is true by definition) and so each of the eight possible sequences of three tosses is equally likely. Hence the results from the previous page on equally likely outcomes give
 - (a) the probability of obtaining two heads followed by a tail is

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Examples of calculation of probabilities

Example 12, continued.

2) the experiment of choosing a person at random and observing their age last birthday, approximate probabilities are (Australian Bureau of Statistics estimates)

(a) the probability that a person is over 100 years old is
 $\mathbb{P}(\{100, 101, 102, \dots\}) = 0.00019;$

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 $\mathbb{P}(\{0, 1, 2, \dots, 17\}) = 0.22.$

How do we allocate probabilities?

Observe that the definition of probability does not give us any idea of what specific probabilities are realistic for any given situation. The preceding examples give two commonly used methods of making this decision.

- Take advantage of any **symmetry** in the problem. This is what we did when we assumed that in tossing a coin, heads and tails are equally likely. It is also what we would do if we assumed that in throwing a die, each face will come up with probability $\frac{1}{6}$, or that in dealing five cards from a standard pack, any specific hand of five cards is as likely as any other.

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- Use experience, or **large amounts of data**. Often this is our only option: it would clearly be unreasonable to expect the age profile of a population to adhere rigidly to any predetermined figures. Indeed, if we are *not* prepared to assume that a coin, die or pack of cards is “fair”, then we have to fall back on looking at long-term relative frequencies.

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- For something like assessing the probabilities of various horses winning the Melbourne Cup, one might combine these methods by initially allocating each horse a winning probability of $\frac{1}{24}$, then adjusting these figures, some upwards, some downwards, by means of **guesswork** (or “judgement”, or “experience” – which both, to a large extent, mean guesswork). However, the results of such a procedure are notoriously unreliable, even when employed by “experts”. This is decidedly not an approach that should be used when dealing with serious issues.

Example 13: The birthday paradox.

What is the probability that if n people are chosen at random, at least two have the same birthday?

Solution. Consider the experiment of choosing n people at random and finding their birthdays. We take the sample space to be

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$$S = \{ (b_1, b_2, \dots, b_n) \mid \text{each } b_k \text{ is a day of the year} \} .$$

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$$S = \{ (b_1, b_2, \dots, b_n) \mid \text{each } b_k \text{ is a day of the year} \} .$$

We are interested in the probability of the event $A \subseteq S$ given by

$$A = \{ (b_1, b_2, \dots, b_n) \in S \mid \text{at least two of } b_1, b_2, \dots, b_n \text{ are the same} \} .$$

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We shall assume that there are 365 possible birthdays, each of them equally probable (of course, neither of these assumptions is actually true); consequently, each of the 365^n elements of S will be taken as equally likely. Now A^c is the set of n -tuples in which no two elements are the same, and the number of such n -tuples is

$$365 \times 364 \times \cdots \times (365 - (n - 1)) .$$

Hence

$$\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \frac{|A^c|}{|S|} = 1 - \frac{365 \times 364 \times \cdots \times (365 - n)}{365^n} .$$

The birthday paradox

Example 13: The birthday paradox, continued.

Here are some calculations of the probability we have just found, in terms of n .

n	10	20	22	23	30	40	50	60
$P(A)$	0.1169	0.4114	0.4757	0.5073	0.7063	0.8912	0.9704	0.9941

In a randomly chosen group of 23 people, there is a greater than 50% chance of two people sharing the same birthday; this is not the case if only 22 people are chosen.



Only 23 people is enough? really?

Problem

Conditional probability. Often we are interested in the probability of an event relative to a reduced sample space.

For example, we might wish to know how the probability of being over 100 differs for males and females. If we take the sample space S to be the whole Australian population and let

$$C = \{ \text{people in } S \text{ over 100 years old} \} , \quad M = \{ \text{males in } S \} , \quad F = \{ \text{females in } S \}$$

then we write

$$\mathbb{P}(C \mid M) ,$$

read “the probability of C **given** M ”, for the probability that a randomly chosen person is over 100, **given that the person is male**; that is, the probability that a randomly chosen male is over 100. ABS estimates for June 2015 indicate that

$$\mathbb{P}(C \mid M) = 0.000086 \quad \text{and} \quad \mathbb{P}(C \mid F) = 0.000286 .$$



Note that this is very different from considering the probability that a randomly chosen 100-year-old is male or female: we have

$$\mathbb{P}(M \mid C) = 0.23 \quad \text{and} \quad \mathbb{P}(F \mid C) = 0.77 .$$

Definition of conditional probability

How did we calculate the probabilities on the previous page? For example, we reasoned that

$$\mathbb{P}(C | F) = \frac{\text{number of 100-year-old females}}{\text{total number of females}} = \frac{|\{100\text{-year-old females}\}|}{|\{\text{females}\}|},$$

which we can write as

$$\mathbb{P}(C | F) = \frac{|C \cap F|}{|F|} = \frac{|C \cap F|/|S|}{|F|/|S|} = \frac{\mathbb{P}(C \cap F)}{\mathbb{P}(F)}.$$

This is the intuition behind the following definition.

Definition (Conditional probability)

Let S be a sample space; let A and B be events in S such that $\mathbb{P}(B) \neq 0$. The **conditional probability** of A **given** B is

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditional probability and Multiplication rule

“The probability of A **given** B ” is by definition $P(A|B) = \frac{P(A \cap B)}{P(B)}$.



TIP!

This is easier to understand and remember in the form (multiply both sides by $P(B)$.)

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A|B) .$$

because it naturally corresponds to the sentence: “The probability that A and B are both true is the probability that B is true times the probability that A is true given that B is true.”

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because it naturally corresponds to the sentence: “The probability that A and B are both true is the probability that B is true times the probability that A is true given that B is true.”

Fact (Multiplication rule)

Given an experiment with sample space S and two events $A, B \subseteq S$ with $P(B) \neq 0$, we have

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A).$$

Proof. This follows immediately from the definition of conditional probability.

As usual, in maths, things are what they should be.

There are various important formulae involving conditional probability. In the following example we hope to show that most of these formulae are just common sense!

Example 14. In the universal set $S = \{\text{students}\}$ consider the subsets A of those who pass and B of those who study hard. Suppose it is known that 90% of those who study hard pass, and 5% of those who don't study hard pass anyway, and that altogether 80% do study hard. Symbolically,

$$\mathbb{P}(A|B) = 0.9, \quad \mathbb{P}(A|B^c) = 0.05, \quad \mathbb{P}(B) = 0.8.$$

a) What proportion study hard **and** pass?

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Answer.

- By common sense, the answer is 90% of the 80% who study hard, i.e. $(0.9)(0.8) = 0.72$.
- Translating common sense into a formula,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B);$$

by symmetry, we also have $\mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A)$.

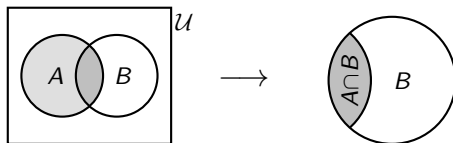
Conditional probability as a change of sample space

“The probability of A given B ” is by definition $P(A|B) = \frac{P(A \cap B)}{P(B)}$.

Conditional probability as a change of sample space

“The probability of A **given** B ” is by definition $P(A|B) = \frac{P(A \cap B)}{P(B)}$.

Diagrammatically, knowing that the event B has occurred can be represented by restricting the first diagram below to the second.



💡 In other words, B is our new sample space.

Our previous example illustrates this:

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$$\mathbb{P}(A|B) = 0.9, \quad \mathbb{P}(A|B^c) = 0.05, \quad \mathbb{P}(B) = 0.8.$$

Comments on Conditional probability.

- It is clear from the conditional probability formula that we must exclude the case $\mathbb{P}(B) = 0$. This should also make sense intuitively: there is not much point in asking about the probability of A , given that B has occurred, if it is impossible for B to have occurred.
- Make sure that you understand the notation $\mathbb{P}(A|B)$ correctly. It is not the probability of “the set $A|B$ ” in the same way that $\mathbb{P}(A \cup B)$ is the probability of the set $A \cup B$. Rather, it is the probability of A , measured not in terms of the whole sample space S , but only under the assumption that B has occurred. In other words, the set B is part of the definition of the function: $P_B(A)$ might be a better notation. In fact, the function which takes the set A and returns the value of $\mathbb{P}(A|B)$ is a probability in the strict sense of the definition.

Lemma (a conditional probability is a probability)

Let P be a probability on S and let B be a subset of S with non-zero probability. Then the function

$$f : \mathcal{P}(S) \rightarrow \mathbb{R} \quad \text{where} \quad f(A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

is a probability on S .

Proof. Check that f satisfies the three axioms on page 18.

Example 14, continued. In the universal set $S = \{\text{students}\}$ consider the subsets A of those who pass and B of those who study hard. Suppose it is known that 90% of those who study hard pass, and 5% of those who don't study hard pass anyway, and that altogether 80% do study hard. Symbolically,

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b) What total proportion pass?

Solution.

- By common sense, 90% of 80% and 5% of 20%, that is,
 $(0.9)(0.8) + (0.05)(0.2) = 0.73$.
- This illustrates the formula

$$\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c),$$

- ... which generalises to

$$\mathbb{P}(A) = \mathbb{P}(A|B_1)\mathbb{P}(B_1) + \mathbb{P}(A|B_2)\mathbb{P}(B_2) + \cdots + \mathbb{P}(A|B_n)\mathbb{P}(B_n),$$

provided $B_1, B_2, \dots, B_n$ are non-empty events which partition the sample space S .

- Alternatively, compile a table of probabilities $A \cap B$ and so on:

	A (pass)	A^c
B (study hard)	0.72	0.08
B^c	0.01	0.19

From this table,
 $\mathbb{P}(A) = 0.72 + 0.01 = 0.73$.

Example 14, continued.

Recall: A = “pass” and B = “study hard”.

$$\mathbb{P}(A|B) = 0.9$$

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$$\mathbb{P}(B) = 0.8 .$$

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- Common sense, using the above table: $\frac{72}{73} = 0.986$.

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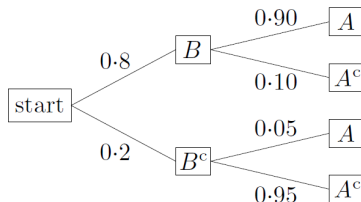
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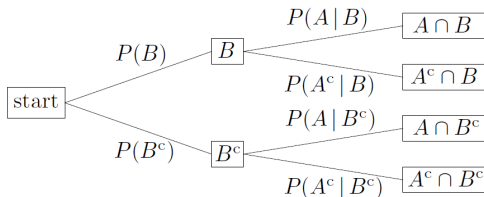
Solution. no. This would mean that the probability of a student passing is the same, regardless of whether we are considering all students, or just those who studied hard. In other words, A and B are independent if $\mathbb{P}(A) = \mathbb{P}(A|B)$. But from our previous calculations, we know that this is not the case.

Tree diagrams

The calculations in the previous example may be illustrated on a **tree diagram**.



Note that strictly speaking, the events in the last column should be labelled not A and A^c but $A \cap B$, $A^c \cap B$ and so on. A more accurately labelled (and generalised) tree diagram would be as follows.



Conditional probability results

Here are general statements of the “common sense” results we saw in our ‘study hard’ and ‘pass’ example .

Theorem (Results on conditional probability)

Let A, B be events in a sample space S .

1. **The multiplication rule:** $\mathbb{P}(A \cap B) = \mathbb{P}(A | B)\mathbb{P}(B) = \mathbb{P}(B | A)\mathbb{P}(A)$.

2. **The total probability rule:** if $B_1, B_2, \dots, B_n$ partition S then

$$\mathbb{P}(A) = \mathbb{P}(A | B_1)\mathbb{P}(B_1) + \mathbb{P}(A | B_2)\mathbb{P}(B_2) + \dots + \mathbb{P}(A | B_n)\mathbb{P}(B_n) .$$

3. **Bayes’ rule:** if $B_1, B_2, \dots, B_n$ partition S then

$$\mathbb{P}(B_k | A) = \frac{\mathbb{P}(A | B_k)\mathbb{P}(B_k)}{\mathbb{P}(A | B_1)\mathbb{P}(B_1) + \mathbb{P}(A | B_2)\mathbb{P}(B_2) + \dots + \mathbb{P}(A | B_n)\mathbb{P}(B_n)} .$$

Proof. The multiplication rule follows from the definition of conditional probability. For the total probability rule, note that $A \cap B_1, A \cap B_2, \dots, A \cap B_n$ partition A , so

$$\mathbb{P}(A) = \mathbb{P}(A \cap B_1) + \mathbb{P}(A \cap B_2) + \dots + \mathbb{P}(A \cap B_n) ;$$

then use the multiplication rule. For Bayes’ rule, we have by definition

$$\mathbb{P}(B_k | A) = \frac{\mathbb{P}(A \cap B_k)}{\mathbb{P}(A)} ,$$

and the result follows from the previous two.

A closer look at Bayes' rule.

Bayes' rule: if $B_1, B_2, \dots, B_n$ partition S then

$$\mathbb{P}(B_k | A) = \frac{\mathbb{P}(A | B_k)\mathbb{P}(B_k)}{\mathbb{P}(A | B_1)\mathbb{P}(B_1) + \mathbb{P}(A | B_2)\mathbb{P}(B_2) + \dots + \mathbb{P}(A | B_n)\mathbb{P}(B_n)}.$$

► *Do I need to memorise Baye's rule?*

NO: it is easy to 'rediscover' once you understand what it says.



In maths, if you understand, you do not need to memorise, or very little.

► *What can Baye's rule do for me?*

Suppose that a number of events $B_1, B_2, \dots, B_n$ partition a sample space, and that their probabilities are known. Suppose also that there is another event A , and that the probability of A given each of the B_k is known. Then

- the total probability rule allows us to find the overall probability that A occurs, even though we do not know which of the B_k occurred;
- Bayes' rule reverses this: if we observe that A has in fact occurred, it allows us to find the probability of each B_k having occurred.

Diagnostics

One of the classic examples of the use of Bayes' rule is its application to medical diagnosis.

Example 16. Imagine a medical test for a serious disease. If the test is taken by a person who has the disease, the test is 98% accurate – that is, there is a 98% chance that the test will show (correctly) that the person has the disease. For a person who does not have the disease, the test is 95% accurate – that is, there is a 95% chance that the test will show that the person does not have the disease. Suppose that the disease occurs in 1% of the population. If a person returns a positive test, what is the probability that they have the disease?

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We should always begin a problem like this by defining a sample space and appropriate events.

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Let the sample space S be the set of all people in the population;
let D be the event that a person has the disease,
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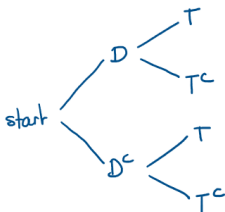
Solution.

Step 1. Setting things up. Sample space S = set of all people in the population; D = 'the person has the disease', T = 'the person tests positive'.

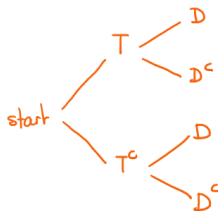
Step 2. Decide which **probability tree** you should use and add on it the probabilities given in the question.



Tree 1: D and D^c first



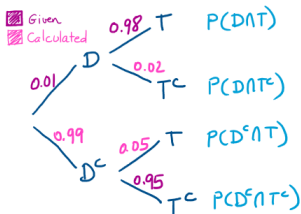
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....continued

Example 16, continued.

Let the sample space S be the set of all people in the population; let D be the event that a person has the disease, and let T be the event that a person tests positive.

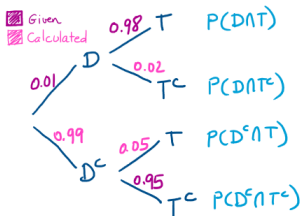


It is given that $\mathbb{P}(T|D) = 0.98$, $\mathbb{P}(T^c|D^c) = 0.95$ and $\mathbb{P}(D) = 0.01$, and we want to know $\mathbb{P}(D|T)$.

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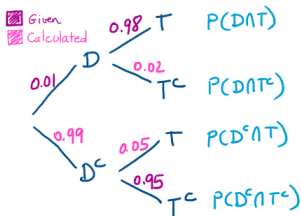
$$\mathbb{P}(T) = \mathbb{P}(T|D)\mathbb{P}(D) + \mathbb{P}(T|D^c)\mathbb{P}(D^c) = (0.98)(0.01) + (0.05)(0.99) = 0.0593,$$

and then by Bayes' rule,

$$\mathbb{P}(D|T) = \frac{\mathbb{P}(T|D)\mathbb{P}(D)}{\mathbb{P}(T)} = \frac{(0.98)(0.01)}{0.0593} = 0.17.$$

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Thus, if a person tests positive, there is a 17% chance that they actually have the disease.

Example 16, continued.

Comment. This is a striking and scarcely believable result!! Since we know that the test delivers the correct result in over 95% of cases, how can it be this bad??

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The answer hinges on the fact that the disease is very rare – only 1% of the population has it. So although virtually all of these people will get a correct positive result, they will be swamped by what are called “false positives” – the (almost) 5% who do not have the disease but still test positive. Therefore, testing positive to the disease means that a person actually has the disease only in a relatively small number of cases.

Also worth noting: this example clearly shows the importance of recognising the difference between $\mathbb{P}(D \mid T)$ and $\mathbb{P}(T \mid D)$.

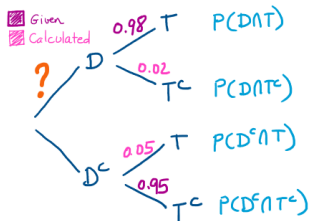
Exercise Assuming the same data, show that if a person returns a negative test, the probability that they have the disease is approximately 0.00021. What are the implications of this for public health policy?

Prevalence of disease



We can alternatively treat the prevalence of the disease in the population as an **unknown**, and use the observed results of the test to estimate this unknown.

Example 17. Continuing the previous example, suppose that the reliability figures for the test remain the same, but that the proportion of the population which has the disease is **unknown**; suppose also that 7% of the population tests positive.



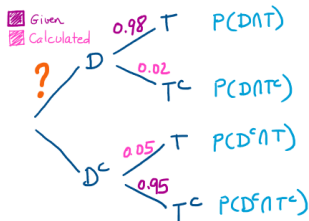
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Solution. We have $\mathbb{P}(T | D) = 0.98$, $\mathbb{P}(T^c | D^c) = 0.95$ and $\mathbb{P}(T) = 0.07$, and we wish to determine $p = \mathbb{P}(D)$.

The total probability rule gives

$$\mathbb{P}(T) = \mathbb{P}(T | D)\mathbb{P}(D) + \mathbb{P}(T | D^c)\mathbb{P}(D^c) = 0.98p + 0.05(1 - p) = 0.05 + 0.93p,$$

and since we know $\mathbb{P}(T) = 0.07$, with a bit of Algebra to make p the subject, we find

$$p = \frac{0.02}{0.93} \approx 0.022.$$

That is, about 2.2% of the population has the disease.

Independence

Recall that in the 'study hard' and 'pass' example we noted, somewhat informally, that events A and B can be regarded as independent if $\mathbb{P}(A) = \mathbb{P}(A|B)$. Using the definition to rewrite this equation leads to the following.

Definition (Independent events)

Events A and B are said to be (statistically) **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) .$$

Comments.

- In this definition we do not need to exclude the case where $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 0$.
- We can express the equation $\mathbb{P}(A) = \mathbb{P}(A|B)$ as

$$\frac{\mathbb{P}(A)}{\mathbb{P}(S)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad \text{or} \quad \frac{\mathbb{P}(B)}{\mathbb{P}(S)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} .$$

Thus, saying that A and B are independent amounts to saying that the probability of the part of A which lies inside B , relative to the probability of B , is the same as the probability of all of A , relative to the whole sample space.



Examples of use of “ A and B are independent $\stackrel{\text{def}}{\iff} \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ ”

Examples 18.

- a) In the ‘study hard’ and ‘pass’ example (A = “pass” and B = “study hard”), we had

$$\mathbb{P}(A) = \frac{73}{100}, \quad \mathbb{P}(B) = \frac{4}{5} \quad \text{and} \quad \mathbb{P}(A \cap B) = \frac{72}{100} \neq \mathbb{P}(A)\mathbb{P}(B),$$

so passing and studying hard are not independent events. (What a surprise. . . !)

- b) In the experiment of tossing three coins, consider the events A : that the first toss is a head, B : that the last toss is a tail, and C : that there are more tails than heads. It is easy to see that all three events have probability $\frac{1}{2}$, since, for example,

$$A = \{ HHH, HHT, HTH, HTT \}.$$

The events A and B are independent since

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{ HHT, HTT \}) = \frac{1}{4} = \mathbb{P}(A)\mathbb{P}(B);$$

this is not surprising, as two events which are “physically independent” may usually be expected to be statistically independent. On the other hand,

$$\mathbb{P}(A \cap C) = \mathbb{P}(\{ HTT \}) = \frac{1}{8} \neq \mathbb{P}(A)\mathbb{P}(C).$$

Again, this is no surprise: obtaining a head on the first toss “makes it less likely” that we will obtain more tails than heads.

For more than two events, the counterpart of 'independence' is 'mutual independence'.

Definition (mutually independent events)

Events $A_1, A_2, \dots, A_n$ are said to be **mutually independent** if for any two or more of these events, the probability of the intersection is equal to the product of the respective probabilities.

Example 19. Is pairwise independence the same as mutually independence?

Roll three dice and denote the results by x, y, z ; consider the sample space $S = \{ (x, y, z) \mid x, y, z = 1, 2, 3, 4, 5, 6 \}$. Define three events

$$A = \{ (x, y, z) \mid x + y \text{ is even} \} ,$$

$$B = \{ (x, y, z) \mid y + z \text{ is even} \} ,$$

$$C = \{ (x, y, z) \mid x + z \text{ is odd} \} .$$

- a) Are A, B and C *pairwise* independent?
- b) Are A, B and C *mutually* independent?

Solution.

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$$C = \{(x, y, z) \mid x + z \text{ is odd}\} .$$

- a) Are A, B and C *pairwise* independent?
- b) Are A, B and C *mutually* independent?

Solution. a) Now

$$A = \{(x, y, z) \mid x, y \text{ are both even}\} \cup \{(x, y, z) \mid x, y \text{ are both odd}\} ,$$

so

$$\mathbb{P}(A) = \frac{|A|}{|S|} = \frac{2 \times 3^2 \times 6}{6^3} = \frac{1}{2} ;$$

and for similar reasons, $\mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2} .$

....continued

Example 19. Is pairwise independence the same as mutually independence?
continued.

Also,

$$A \cap B = \{ (x, y, z) \mid x, y, z \text{ are all even} \} \cup \{ (x, y, z) \mid x, y, z \text{ are all odd} \}$$

and therefore

$$\mathbb{P}(A \cap B) =$$

Example 19. Is pairwise independence the same as mutually independence?
continued.

Also,

$$A \cap B = \{ (x, y, z) \mid x, y, z \text{ are all even} \} \cup \{ (x, y, z) \mid x, y, z \text{ are all odd} \}$$

and therefore

$$\mathbb{P}(A \cap B) = \frac{2 \times 3^3}{6^3} = \frac{1}{4} = \mathbb{P}(A)\mathbb{P}(B) ;$$

so A and B are independent.

Example 19. Is pairwise independence the same as mutually independence?
continued.

Also,

$$A \cap B = \{ (x, y, z) \mid x, y, z \text{ are all even} \} \cup \{ (x, y, z) \mid x, y, z \text{ are all odd} \}$$

and therefore

$$\mathbb{P}(A \cap B) = \frac{2 \times 3^3}{6^3} = \frac{1}{4} = \mathbb{P}(A)\mathbb{P}(B) ;$$

so A and B are independent. In the same way, we can show that B and C are independent, as are A and C .

b)

Example 19. Is pairwise independence the same as mutually independence?
continued.

Also,

$$A \cap B = \{ (x, y, z) \mid x, y, z \text{ are all even} \} \cup \{ (x, y, z) \mid x, y, z \text{ are all odd} \}$$

and therefore

$$\mathbb{P}(A \cap B) = \frac{2 \times 3^3}{6^3} = \frac{1}{4} = \mathbb{P}(A)\mathbb{P}(B) ;$$

so A and B are independent. In the same way, we can show that B and C are independent, as are A and C .

b) However, if $x + y$ and $y + z$ are even then $x + z$ must also be even; so

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\emptyset) = 0 \neq \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) .$$

Therefore, A, B, C are not mutually independent.

Comment. We can think of the situation intuitively as follows:

- any two of these three events are independent; that is, the fact that any single event has occurred does not change the probability of any other single event occurring;
- however, all three are *not* mutually independent; that is, the fact that two of them have occurred *does* change the probability of the third occurring.

Independence of combinations

Example 20. Show that if A, B are independent, then

- a) A^c and B are independent. *Let's do only (a) during the lecture*
- b) A and B^c are independent. *Left as an exercise.*
- c) A^c and B^c are independent. *Left as an exercise.*

Proof.



What about three events? Challenge accepted!

Independence of combinations

Example 21. Show that if A, B, C are mutually independent, then $A^c \cup B$ and C are (mutually) independent.

Proof. We can see from a Venn diagram, or otherwise, that $(A^c \cup B) \cap C$ is partitioned by $A^c \cap C$ and $A \cap B \cap C$; so

$$\mathbb{P}((A^c \cup B) \cap C) = \mathbb{P}(A^c \cap C) + \mathbb{P}(A \cap B \cap C).$$

By similar arguments, we have

$$\mathbb{P}(C) = \mathbb{P}(A^c \cap C) + \mathbb{P}(A \cap C) \quad \text{and} \quad \mathbb{P}(A^c \cup B) = \mathbb{P}(A^c) + \mathbb{P}(A \cap B).$$

Using the mutual independence of A, B, C , we have

$$\begin{aligned}\mathbb{P}((A^c \cup B) \cap C) &= \mathbb{P}(C) - \mathbb{P}(A \cap C) + \mathbb{P}(A \cap B \cap C) \\ &= \mathbb{P}(C) - \mathbb{P}(A)\mathbb{P}(C) + \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) \\ &= (1 - \mathbb{P}(A) + \mathbb{P}(A)\mathbb{P}(B))\mathbb{P}(C) \\ &= (\mathbb{P}(A^c) + \mathbb{P}(A \cap B))\mathbb{P}(C) = \mathbb{P}(A^c \cup B)\mathbb{P}(C).\end{aligned}$$

Therefore $A^c \cup B$ and C are independent.

That was a bit painful! Is there a way to avoid writing such a proof? Like a theorem which would ensure that $A^c \cup B$ and C are independent as soon as A, B, C are mutually independent?

Independence of combinations

Example 21. Show that if A, B, C are mutually independent, then $A^c \cup B$ and C are (mutually) independent.

Proof. We can see from a Venn diagram, or otherwise, that $(A^c \cup B) \cap C$ is partitioned by $A^c \cap C$ and $A \cap B \cap C$; so

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By similar arguments, we have

$$\mathbb{P}(C) = \mathbb{P}(A^c \cap C) + \mathbb{P}(A \cap C) \quad \text{and} \quad \mathbb{P}(A^c \cup B) = \mathbb{P}(A^c) + \mathbb{P}(A \cap B) .$$

Using the mutual independence of A, B, C , we have

$$\begin{aligned} \mathbb{P}((A^c \cup B) \cap C) &= \mathbb{P}(C) - \mathbb{P}(A \cap C) + \mathbb{P}(A \cap B \cap C) \\ &= \mathbb{P}(C) - \mathbb{P}(A)\mathbb{P}(C) + \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) \\ &= (1 - \mathbb{P}(A) + \mathbb{P}(A)\mathbb{P}(B))\mathbb{P}(C) \\ &= (\mathbb{P}(A^c) + \mathbb{P}(A \cap B))\mathbb{P}(C) = \mathbb{P}(A^c \cup B)\mathbb{P}(C) . \end{aligned}$$

Therefore $A^c \cup B$ and C are independent.

That was a bit painful! Is there a way to avoid writing such a proof? Like a theorem which would ensure that $A^c \cup B$ and C are independent as soon as A, B, C are mutually independent? YES

Independence of combinations

Problem

If we know that A , B and C are mutually independent, wouldn't it be nice to be able to say without having to prove it that A and $B^c \cup C$ must also be independent?

Theorem (Independence of combinations)

Suppose that the events $A_1, A_2, \dots, A_n$ are mutually independent, and that $B_1, B_2, \dots, B_m$ are formed from (some or all of) the sets A_k by means of unions, intersections and complements. Suppose also that no set A_k occurs more than once, either in the same B_i or in two different B_i, B_j . Then $B_1, B_2, \dots, B_m$ are mutually independent.

Proof. Omitted – an intricate and messy induction.

Let's practice using this theorem.



Example 22. Suppose A, B, C and D are mutually independent events. Does the *Independence of combinations* theorem imply that

- a) the events $A \cap B^c$, C and D^c are mutually independent? ☐ YES ☐ NO
- b) the events B^c and C are independent? ☐ YES ☐ NO
- c) the events $A \cap B^c$ and $A \cup C$ are independent? ☐ YES ☐ NO
- d) the events $A \cap B^c$ and $C \cup D$ are independent? ☐ YES ☐ NO